

Online Resource 1 for “What can country panels tell us about broadband price responsiveness? Evidence from the European Union and the Eastern Partnership, 2010–2024”

Abstract

This Online Resource reports additional estimates and the data-construction details referred to in the article. All estimates can be reproduced from public data with the code in the replication package.

Additional tables

Table S1 First-difference estimates, 2010–2019: pooled, EU and EaP coefficients under alternative choices

Specification	Pooled		EU		EaP		Obs.
	Coef.	SE	Coef.	SE	Coef.	SE	
Baseline	−0.07*	(0.04)	−0.01	(0.02)	−0.28***	(0.08)	240
Price in PPP dollars	−0.03	(0.03)	−0.02	(0.02)	−0.10	(0.23)	233
Price in current US dollars	−0.06	(0.04)	−0.02	(0.02)	−0.23*	(0.12)	242
GDP only	−0.07*	(0.04)	−0.01	(0.02)	−0.27***	(0.07)	240
+ servers and R&D	−0.06	(0.04)	0.00	(0.02)	−0.25***	(0.09)	240
+ mobile basket (2014–19)	−0.01	(0.01)	−0.01	(0.01)	−0.02	(0.04)	140
Internet users outcome	−0.08***	(0.03)	−0.03*	(0.01)	−0.27***	(0.03)	240
Incl. repeated 2019, 2017–18	−0.06*	(0.03)	−0.01	(0.01)	−0.27***	(0.08)	295
Log count of subscriptions	−0.06*	(0.03)	−0.01	(0.01)	−0.26***	(0.08)	295
Sample ending in 2017	−0.09*	(0.05)	−0.01	(0.02)	−0.28***	(0.08)	229
Log-odds, $K = 55$	−0.08*	(0.04)	−0.02	(0.02)	−0.31***	(0.09)	240
Log-odds, $K = 70$	−0.08*	(0.04)	−0.01	(0.02)	−0.30***	(0.08)	240
Log-odds, $K = 100$	−0.07*	(0.04)	−0.01	(0.02)	−0.29***	(0.08)	240
Driscoll–Kraay SEs			−0.01	(0.01)	−0.28***	(0.08)	240

First differences with year effects and core controls; clustered standard errors unless stated. Log-odds rows use $\ln[s/(K - s)]$ as the outcome. The mobile-broadband basket is available from 2013, which restricts the differenced sample to 2014–2019. Driscoll–Kraay: Bartlett kernel, bandwidth 3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S2 Functional form and headroom, first differences, 2010–2019

<i>Panel A. Implied elasticity at group means</i>				
Form	EU	p	EaP	p
Linear (s on price)	−0.00	0.852	−0.04	0.277
Log-linear ($\ln s$ on price)	0.00	0.952	−0.21	0.000
Linear-log (s on $\ln$ price)	−0.01	0.507	−0.12	0.114
<i>Panel B. Price coefficient varying with headroom $h_{i,t-1} = 1 - s_{i,t-1}/70$ (centred)</i>				
	2010–2019		2010–2024	
$\Delta \ln p$	−0.02 (0.02)		−0.04*** (0.02)	
$\Delta \ln p \times h$	−0.69* (0.35)		−0.49* (0.28)	
$\Delta \ln p \times \text{post-2019}$			0.02 (0.02)	
<i>Adding the EaP interaction (2010–2019)</i>				
$\Delta \ln p; \times \text{EaP}; \times h$	−0.00 (0.02);	−0.23*** (0.05);	−0.19 (0.29)	

Panel A: elasticities evaluated at group mean price and penetration; p -values refer to the underlying coefficient (EaP: sum of main and interaction terms). Panel B: clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S3 Price $\times$ post-2019: sample restrictions

Sample	TWFE levels		First differences		Obs. (FD)
	Price	Price $\times$ post	Price	Price $\times$ post	
Full sample	−0.22*** (0.05)	0.30*** (0.07)	−0.07** (0.03)	0.05** (0.02)	382
Excl. UKR 2022–24, BLR 2020–24	−0.23*** (0.05)	0.29*** (0.07)	−0.06* (0.03)	0.05** (0.02)	375
Ending in 2021	−0.21*** (0.06)	0.27*** (0.07)	−0.07* (0.04)	0.06 (0.03)	284

Core controls; country and year effects (levels) or year effects (first differences); clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S4 Hausman-type instrumental-variable estimates, pooled

Estimator	Instrument	Coef. (SE)	First-stage F	Anderson–Rubin 95% set
Levels, 2010–2019	sub-regional	−1.31 (0.46)	12.0	[−3.2, −0.6]
Levels, 2010–2019	3 nearest capitals	−2.18 (1.51)	2.5	[−6.0, −0.4] (grid edge)
FD, 2010–2019	sub-regional	1.81 (2.30)	1.0	—
FD, 2010–2019	nearest capitals	0.96 (0.93)	2.1	—
FD, 2010–2019	both	1.03 (1.01)	2.1	—
FD, 2010–2024	sub-regional	61.40 (2072.53)	0.0	—
FD, 2010–2024	nearest capitals	0.95 (0.87)	2.1	—
FD, 2010–2024	both	0.49 (0.49)	2.4	—

Instruments are averages of log prices of other sample countries in the same year: the leave-one-out mean of the country’s sub-region, or the inverse-distance-weighted mean of the three countries with the nearest capitals. Levels regressions are two-way demeaned; FD regressions include year effects. Core controls; standard errors clustered by country. The Anderson–Rubin (AR) set was computed on a grid over [−6, 4]. Sub-regions: Nordic (DNK, FIN, SWE), Baltic (EST, LVA, LTU), Western (AUT, BEL, DEU, FRA, IRL, LUX, NLD), Southern (CYP, ESP, GRC, ITA, MLT, PRT), Central (CZE, HUN, POL, SVK), South-Eastern (BGR, HRV, ROU, SVN), South Caucasus (ARM, AZE, GEO), Eastern (BLR, MDA, UKR). The first-difference first stages are too weak for the estimates to be informative.

Table S5 Country-specific first-difference slopes (2010–2019) and weights in the levels regression

Country	Slope	SE	Obs.	TWFE weight	Country	Slope	SE	Obs.	TWFE weight
LTU	-0.27	0.18	7	0.015	HRV	0.01	0.01	8	0.063
CZE	-0.20	0.06	7	0.021	LUX	0.01	0.08	7	0.005
PRT	-0.17	0.15	7	0.011	BEL	0.01	0.03	7	0.021
FIN	-0.06	0.08	7	0.011	DNK	0.02	0.07	7	0.004
BGR	-0.04	0.10	7	0.010	MLT	0.02	0.02	7	0.038
DEU	-0.03	0.03	7	0.005	ESP	0.03	0.09	8	0.041
NLD	-0.03	0.04	8	0.010	HUN	0.05	0.10	7	0.023
GRC	-0.02	0.06	7	0.022	SWE	0.08	0.09	7	0.023
POL	-0.02	0.08	7	0.043	ROU	0.12	0.45	5	0.012
CYP	-0.02	0.02	7	0.027	EST	0.17	0.13	7	0.014
AUT	-0.02	0.05	8	0.019	AZE	-0.75	0.34	7	0.025
IRL	-0.02	0.02	7	0.055	ARM	-0.41	0.16	7	0.155
LVA	-0.00	0.04	8	0.044	MDA	-0.41	0.23	7	0.009
SVN	-0.00	0.03	8	0.021	BLR	-0.21	0.11	7	0.084
SVK	0.01	0.05	8	0.016	UKR	-0.05	0.21	8	0.031
ITA	0.01	0.04	8	0.081	GEO	0.67	0.42	8	0.020
FRA	0.01	0.03	8	0.021					

Slopes: regression of the annual change in log subscriptions per 100 on the change in log price and log GDP per capita with a constant, by country; conventional standard errors. EaP countries in bold. TWFE weight: country's share of the sum of squared residualised prices in the 2010–2019 levels regression (price residualised on country and year effects and core controls); the six EaP countries account for 32%.

Table S6 Data construction: flags, imputations and sample composition

Item	Treatment
ITU basket definition	1 GB minimum (2010–2017), 5 GB minimum (2018–2024); the 2017–2018 change is never used in differenced regressions.
2019 prices equal to 2018 in all three units	22 countries (ARM, AZE, BEL, BGR, BLR, CYP, CZE, DEU, DNK, EST, FIN, GRC, HUN, IRL, LTU, LUX, MDA, MLT, POL, PRT, ROU, SWE); set to missing.
Missing in source	Romania 2013 price; Bulgaria 2024 subscriptions; left missing.
Mobile-broadband basket	Chain of ITU data-only baskets: postpaid computer-based 1 GB (2013–2017), 1.5 GB (2018–2020), 2 GB (2021–2024).
Tertiary enrolment	9 interior values interpolated, 21 end values set to nearest observation.
R&D expenditure (robustness only)	30 end values set to nearest observation.
Population density	2024 values computed from population and land area.
Night-time lights	Harmonised DMSP (to 2013) and simulated VIIRS (from 2014) radiance summed over Natural Earth national boundaries (European France only); the 2013–2014 change is excluded.
First-difference observations by year	2011: 33, 2012: 33, 2013: 32, 2014: 32, 2015: 33, 2016: 33, 2017: 33, 2019: 11, 2020: 11, 2021: 33, 2022: 33, 2023: 33, 2024: 32.
Earlier version of this panel	Used $\ln(1+p)$ instead of $\ln p$, forward-filled missing prices and subscriptions, and averaged different mobile baskets; all corrected here.

Table S7 EaP first-difference coefficient by year and identification checks, 2010–2019

<i>Panel A. EaP coefficient by year (common EU slope, year effects, core controls)</i>			
Year	Coef.	SE	EaP countries with data
2011	−0.75	(0.20)	6
2012	−0.07	(0.23)	6
2013	−0.61	(0.26)	6
2014	−0.28	(0.11)	6
2015	−0.58	(0.22)	6
2016	−0.24	(0.14)	6
2017	−0.08	(0.06)	6
2019	0.80	(0.19)	2
<i>Panel B. Alternative fixed effects in first differences</i>			
Specification	EU	EaP	WCR p (difference)
Year effects (baseline)	−0.01	−0.28***	0.037
Year and country effects	−0.00	−0.17**	0.133
Group-specific year effects	−0.00	0.13	0.523

Panel A: coefficient on the change in log price for EaP countries in each year, from one regression in which the EaP slope is year-specific and the EU slope common; 2018 is omitted because of the basket revision, and in 2019 only two EaP countries report a new price. Panel B: first-difference models of Table 4 in the article; with group-specific year effects the EaP coefficient is identified only from differences among EaP countries within a year. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

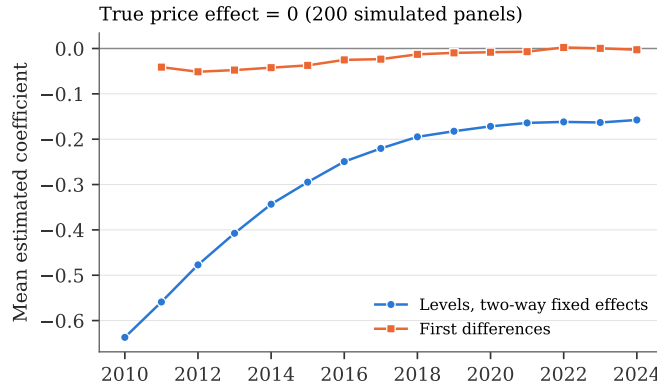


Fig. S1 Simulated year-specific coefficients when the true price coefficient is zero. 33 countries follow logistic adoption paths (ceilings 35–50 per 100, speeds 0.25–0.45); 27 have diffusion midpoints before the sample (mean −4 years), six start late (mean midpoint in year 3) with affordability ratios about 1.3 log points higher that fall with the stage of diffusion. Idiosyncratic noise: 0.03 in log adoption, 0.15 in log price. Lines show means over 200 panels of the year-specific coefficients from the levels regression with country and year effects and from first differences with year effects. The pooled coefficients average −0.48 (levels) and −0.02 (first differences)